

MARKET STRUCTURES (CONTINUED) & WELFARE ECONOMICS

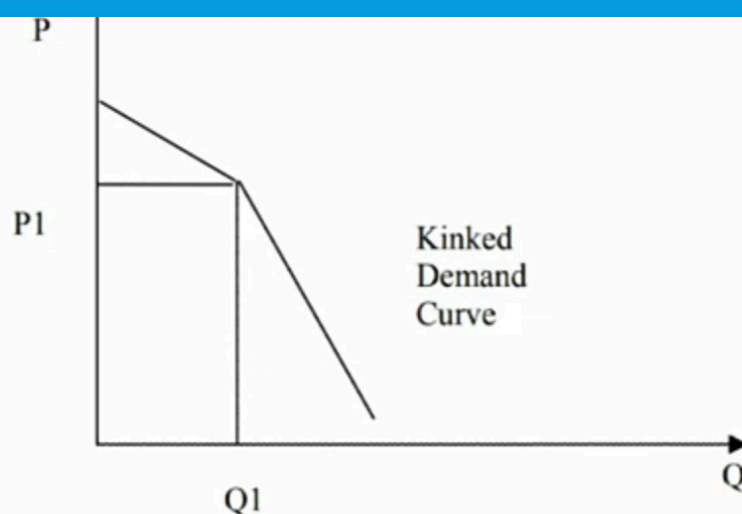
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PRICE STABILITY IN NON-COLLUSIVE OLIGOPOLIES: KINKED DEMAND CURVE

□ A kinked demand curve explains the “stickiness” of the prices in oligopolistic markets. The theory of kinked demand curve rests on the two assumptions that if one firm raises prices, no one else will raise their prices and so the firm will face declining revenues (elastic demand). However if one firm lowered its price, everyone else would lower their prices as well and everyone’s revenues, including the first firm’s revenues would fall (inelastic demand).

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A DEMAND CURVE WITH TWO DISTINCT
SEGMENTS WHICH HAVE DIFFERENT
SLOPES THAT JOINS TO FORM A CORNER O

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The two segments are:

- (1) A relatively more elastic segment for price increases
 - (2) A relatively less elastic segment for price decreases
- The relative elasticities of these two segments are based on the interdependent decision-making of oligopolistic firms.

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□ **Non Price Competition:** Non price competition means competition amongst the firms based on factors other than price, e.g. advertising expenditures.

□ **Oligopoly & public interests:** In oligopoly, firms are able to earn super normal profits. This is also the feature of monopoly. But this is not the feature of perfect competition & monopolistic competition. Firms can use their profits in cost minimization techniques

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WELFARE ECONOMICS

□ It is a branch of economics dealing with normative issues (i.e., what should be). Welfare economics is a branch of economics that uses microeconomic techniques to simultaneously determine allocative efficiency within an economy and the income distribution associated with it. It analyzes social welfare in terms of economic activities of the individuals that comprise the theoretical society considered.

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THE MARGINAL PRIVATE COST OF ADVERTISING

□ The marginal private cost of advertising is the cost of every additional TV commercial or newspaper advertisement that a firm has to bear. However, this does not include the nuisance cost that such advertisements sometimes cause to viewers of television or readers of newspapers. If firms incorporated these costs into their calculations, they would do less advertising. Concerns such as these fall into the realm of welfare economics.

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MARGINAL SOCIAL COST

□ Social cost (benefit) means the cost (benefit) – may not be in monetary terms – that is borne by (accrues to) society on the whole. The private cost (benefit) of any individual entity (firm or consumer) is subsumed in the social cost (benefit) to society, but obviously not vice versa. The marginal social cost = marginal private costs that the firm incurs + any other costs that is borne by the society because of the production of additional good

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THE CONCEPT OF EXTERNALITY

Formally, an externality exists when the production or consumption of a good directly affects businesses or consumers not involved in buying and selling it and when those spillover effects are not fully reflected in market prices.

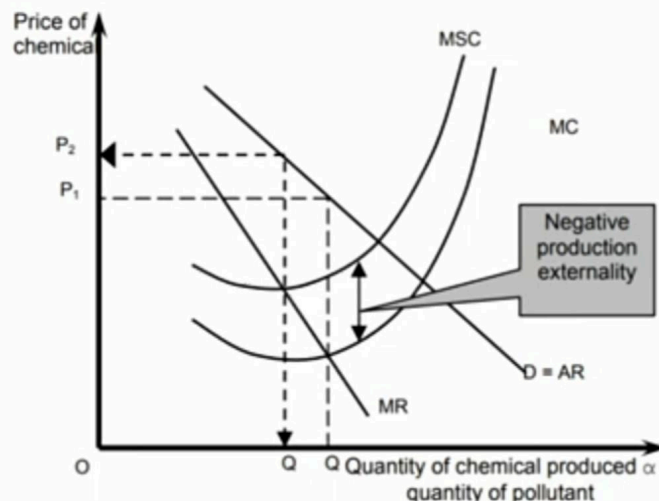
A positive (negative) externality arises from the beneficial (harmful) spillover effect of production or consumption for society. If the externality is a result of private production decisions, it is called a production externality. If it is caused by private consumption decisions, it is called a consumption externality.

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OPTIMAL LEVEL OF PRODUCTION

A socially optimal level of production of a good means the level of production at which the externality is fully internalized, i.e. the equilibrium price and quantity are determined at the intersection of the marginal "social" benefit curves and marginal "social" cost curves, and NOT the intersection of marginal private benefit (demand) curves and marginal private cost (supply) curves.

Tax:

A tax (subsidy) raises (reduces) prices by shifting the supply curve vertically upwards (downwards).

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MARKET FAILURE

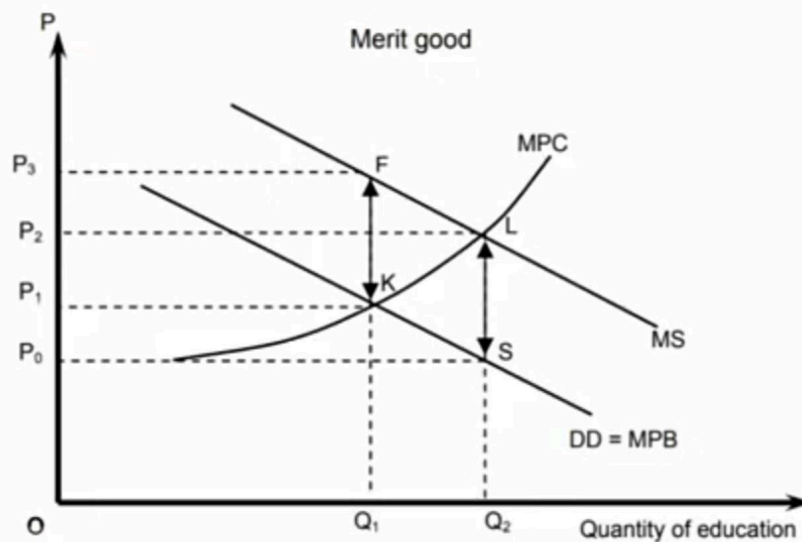
❑ Market failure is an imperfection in the price system that prevents an efficient allocation of resources.

❑ MERIT GOOD:

❑ Merit good is a good which Govt like people to consume more. A merit good in economics is a commodity which is judged that an individual or society should have on the basis of a norm other than respecting consumer preferences. Examples include food stamps, health care, and subsidized housing. If Govt found positive consumption externality then there would be the case of subsidy. For example, Govt wants to encourage people for more education and it thinks that marginal social benefit of education is high than the marginal private benefit so it gives scholarships and subsidies students and education sector.

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PUBLIC GOOD

A public good is one whose benefits are indivisibly spread among the entire community, whether or not particular individuals desire to consume the good or not. There are two characteristics which give rise to public goods: non-rival ness (one person's use or consumption of the good does not reduce the ability of another to use it; e.g. air) and non-excludability (it is not possible to exclude anyone from the consumption of the good; e.g. national defense).

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- In economics, a public good is a good that is non-rival and non-excludable. This means that consumption of the good by one individual does not reduce the amount of the good available for consumption by others; and no one can be effectively excluded from using that good. Common examples of public goods include: defense and law enforcement (including the system of property rights), public fireworks, lighthouses, clean air and other environmental goods, and information goods, such as software development, authorship, and invention.

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WELFARE ECONOMICS (CONTINUED)

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THE MARKET FOR FACTORS OF PRODUCTION

The circular flow of income and expenditure shows the flow of goods and factors between households and firms. Firms are the demanders of the factors and households are the suppliers of the factors.

The Demand for Factor of Production:

The demand for factors of production (like labor) is a derived demand, because it is "derived" from the goods market

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LEISURE:

Leisure is the time not used for working, or earning wages.

The Marginal Disutility of Work (MDUW):

As the supply of hours of labor increases, wage rate should also be increased. The relationship between hours provided by labor and wage rate is positive. And the curve for labor supply curve is positively sloped. But this curve can also be negatively sloped due to the marginal disutility of working hours. The marginal disutility of work (MDUW) means the negative impact on the working of laborer for one additional unit of time. The MDUW curve defines supply curve for labor.

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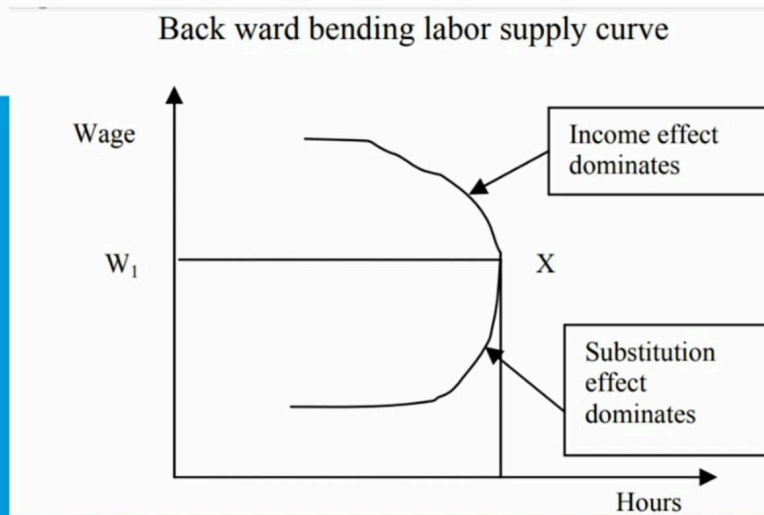
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THE OPPORTUNITY COST:

The opportunity cost of working is leisure (and vice versa) that the worker could have enjoyed during that time had he not been working.

Supply and Demand Curve for Labor:

The labour supply curve may bend backwards above a certain wage rate as the income effect of higher wages dominates the substitution effect of higher wages.



THE VALUE OF MARGINAL PRODUCT OF LABOR (VMPL):

The value of marginal productivity of labor can be represented by the following formula:

The value of marginal product of labour (VMPL) = $MPPL \times P_i$

RENT:

A rent is a periodic payment as a reward for hiring the factor of production for that period, where the purchase price of capital is the "value" of owning that capital for its entire life.

The Net Present Value (NPV) and Discounting:

Decisions about purchasing capital or land are often made on the basis of the net present value (NPV) associated with the decision. The NPV of an asset is the discounted value of the net returns that the asset generates over a period of time plus the discounted value of its disposal value at the end of the period minus the initial purchase cost.

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THE ECONOMICS OF INFORMATION PRODUCTS

- The economics of information products or (internet products) involves studying how economic principles apply to the production, distribution and consumption of these products. Information economics or the economics of information is a branch of microeconomic theory that studies how information affects an economy and economic decisions.
- The internet has reduced the marginal cost of distributing information to zero, as once a product is launched on the web, any number of potential customers can access/view it without any additional cost to the producer of the information.

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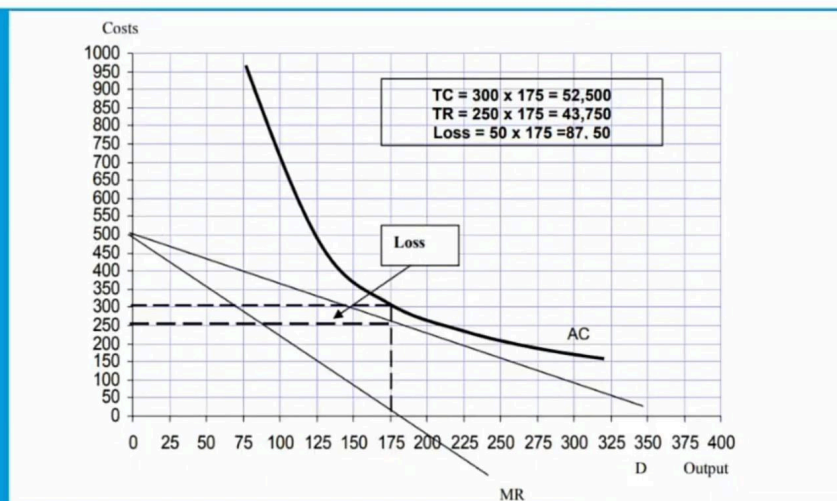
A typical information product

Online economics course			
Q	TC	MC	AC
50	50,000		1000
100	50,000	0	500
150	50,000	0	330
200	50,000	0	250
250	50,000	0	200
300	50,000	0	166

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A TYPICAL INFORMATION PRODUCT PROFIT OR LOSS



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INTRODUCTION TO MACROECONOMICS

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MACROECONOMICS

As a subject, macroeconomics only began to be taught in colleges and universities in the 1940s after the influence of a very influential British economist, John Maynard Keynes who believed the macro economy (with its associated variables) deserved to be understood and analyzed in its own right, and not just as an aggregation of the various micro-markets, as was believed earlier.

Macroeconomics is a branch of economics that deals with the performance, structure, and behavior of a national economy as a whole. The variables of interest change from the price, demand or supply of a particular product to the economy-wide price level, aggregate demand and aggregate supply.

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AGGREGATE DEMAND (AD)

Aggregate demand (AD) is the total planned or desired spending (expenditure) in the economy during a given period. AD is the sum of consumption, investment, government spending and net exports (i.e. exports minus imports), and is inversely related to the aggregate price level through the wealth, interest rate and international purchasing power effects

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AGGREGATE SUPPLY (AS)

Aggregate supply (AS) is the total value of goods and services that all the firms in the economy would and can willingly produce in a given time period. Aggregate supply is a function of available inputs, technology and the price level. It slopes upward in P-Output space but the exact slope depends whether the economy is operating at below full employment (flat) or full employment (steep).

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CLASSICAL ECONOMICS

Classical economics is widely regarded as the first modern school of economic thought. Its major developers include Adam Smith, David Ricardo, Thomas Malthus and John Stuart Mill. Sometimes the definition of classical economics is expanded to include William Petty, Johann Heinrich von Thünen, and Karl Marx. Classical economists were the earliest brand of economists the world knew. They were essentially micro-economists who believed the macro economy was an uninteresting aggregation of individual (or micro) markets, and any problem at the macro level was necessarily a symptom of some micro level problem.

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OPTIMAL ROLE OF GOVERNMENT UNDER CLASSICAL ECONOMICS

The optimal role for the government under classical economics was one of laissez-faire. They believed that if the prices of goods, services and factors were allowed to be determined by the free operation of the forces of demand and supply (i.e. the price mechanism) the best possible outcome for resource allocation would obtain. In other words the economy would be at the full employment level, and it would not be possible to improve that situation through government intervention.

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Before 1930, there was no concept of macroeconomics. There were number of events happened during 1920-30s that necessitated the need of macroeconomics. Until the 1930s, most economic analysis did not separate out individual behavior from aggregate behavior. With the Great Depression of the 1930s and the development of the concept of national income and product statistics, the field of macroeconomics began to expand.

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THE CONCEPT OF INVISIBLE HAND

Invisible hand was a concept introduced by Adam Smith in 1776 to describe the paradox of laissez-faire market economy. The invisible hand doctrine holds that, with each participant pursuing his or her own private interest, a market system nevertheless works to the benefits of all as though a benevolent invisible hand were directing the whole process. According to Adam Smith, invisible hand of the market operates therefore the market mechanism is the best model if the economy wants to operate at a high level of efficiency. Since the classical economists believed on perfectly competitive markets, so according to them shortages & surpluses are temporary phenomena when markets decide about the price, market would automatically clear these shortages & surpluses through price mechanism.

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wage rate goes down, firms will demand more labor due to lower wage rates, so in this way, supply of labor creates its own demand of labor.

Invisible hand is the term used by Adam Smith to describe the natural force that guides free market capitalism through competition for scarce resources. According to Adam Smith, in a free market each participant will try to maximize self-interest, and the interaction of market participants, leading to exchange of goods and services, enables each participant to be better off than when simply producing for him/her. He further said that in a free market, no regulation of any type would be needed to ensure that the mutually beneficial exchange of goods and services took place, since this "invisible hand" would guide market participants to trade in the most mutually beneficial manner.

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An important law the Classical subscribed to, which assumed particular importance in the context of the Great Depression, was Say's law: "supply creates its own demand." The implication of this was that involuntary unemployment (people being unemployed against their wishes) was a temporary phenomenon as the excess supply of labour would cause wages to fall thereby prompting firms to demand more labour. If there was persistent unemployment, it was voluntary, i.e. workers themselves preferred to remain unemployed.

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THE CLASSICAL VIEWS ABOUT GREAT DEPRESSION

The Classical' reading of the three problems of the Great Depression, i.e. low investment, high unemployment and low output, was as follows:

- a. Investment was low because the interest rate was too high in the loanable funds market. Policy recommendation: savings be increased to lower the interest rate and boost investment.
- b. Unemployment was high because of obstructions to the free market mechanism in the labour market which were preventing wages from falling to the market clearing level. Policy recommendation: these obstructions: benefit payments to unemployed, taxes on income and trade unions be eliminated.

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FAILURE OF THE CLASSICAL MODEL

After 1930, the classical model failed. Solution of classical economist was not found reasonable to solve the world crises prevailed at that time. The Great Depression was the longest and severest recession the world has ever seen. It struck North America and Europe in the late 1920s after the Wall Street crash of 1929 (and following the earlier hyperinflation in Germany, and the formation of the Soviet Union) and lasted till the mid 1930s. It was characterized by persistent high unemployment, low investment by firms and falling prices of goods, services and factors. Hyperinflation is inflation at extremely high rates (say 1000, 1 million, or even 1 billion percent a year).

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At that time, unemployment rate went upto 25% in 1933 in USA and Western Europe. The classical model failed because they did not give satisfactory solution to all these problem. They kept on insisting on old doctrine. They focused on the removal of impediments of free market economy

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KEYNES AND THE ORIGINS OF MODERN MACRO ECONOMICS

Keynesian economics also called Keynesianism or Keynesian Theory is an economic theory based on the ideas of the 20th-century British economist John Maynard Keynes. Keynesian economics promotes a mixed economy, in which both the state and the private sector are considered to play an important role. He argued that government policies could be used to promote demand at a macro level, to fight high unemployment and deflation of the sort seen during the 1930s. Keynes gave the reasons of great depression and also suggested the policy advice on how Govt can rectify the situation.

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THE KEYNESIANS' VIEWS ABOUT GREAT DEPRESSION

Keynes' view on the causes of the Great Depression and what needed to be done was very different. He believed that there were overarching problems of low demand and static pessimistic expectations that needed to be addressed rather than disequilibria in the loanable funds, labour and goods markets. In particular, he maintained that:

a. Low investment was because of firms' bearish expectations about their ability to sell the products they produced. Firms needed to see that the potential buyers of their goods had the money and the willingness to buy goods before they could be convinced to undertake more production thereof. Therefore, higher savings, which would lead to low consumption expenditure on goods and services would not increase but decrease investment by reinforcing firms' bearish expectations about their ability to sell their product. Policy recommendation: households should be convinced to increase consumption and reduce saving.

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INTRODUCTION TO MACROECONOMICS (CONTINUED)

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DIFFERENCES BETWEEN CLASSICAL AND KEYNESIAN ECONOMICS

We can see the major differences between classical and Keynesian economics in three ways:

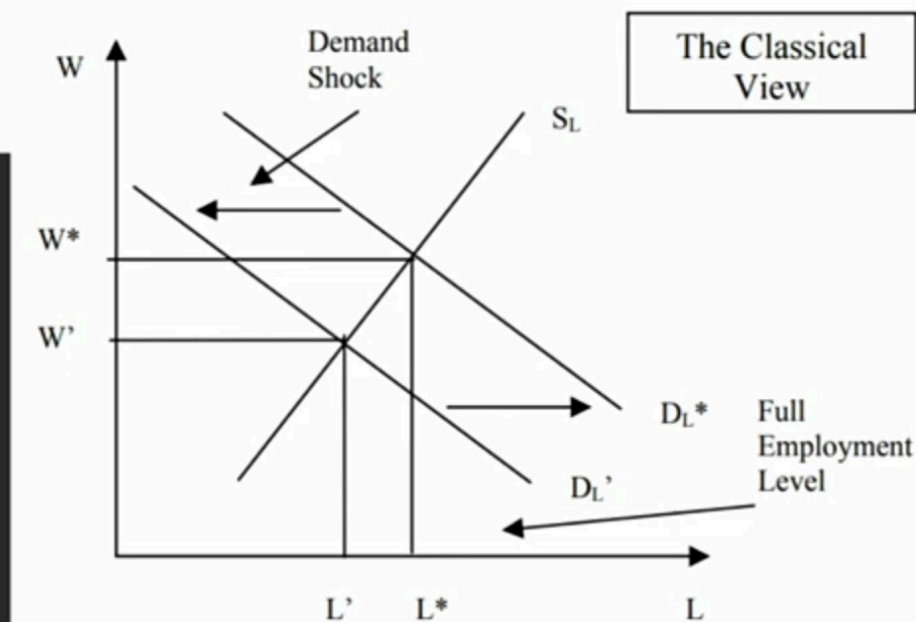
1. Labor market
2. Market for loanable funds
3. Aggregate demand and supply

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1- (A) LABOR MARKET: THE CLASSICAL VIEW

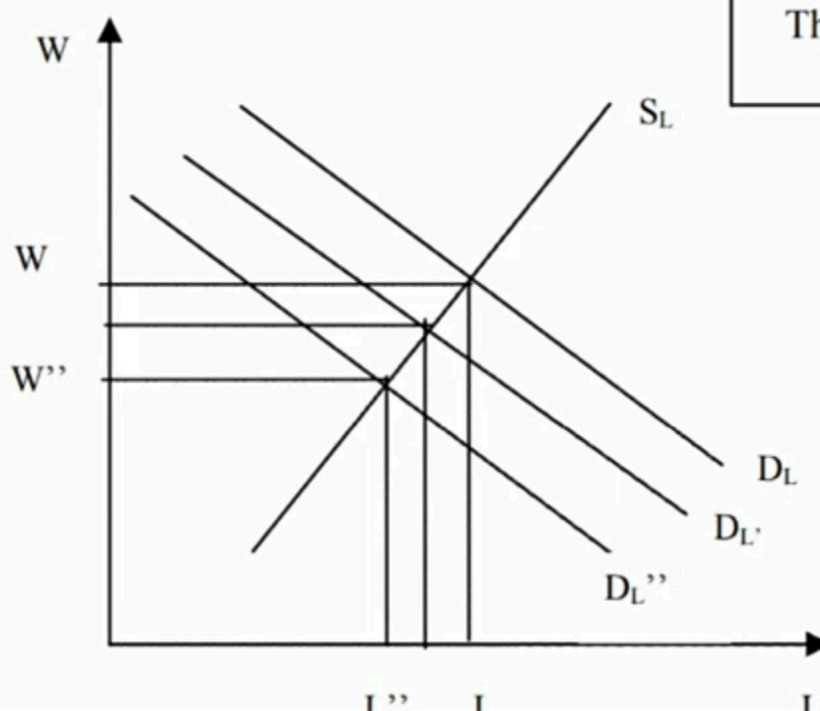
If there is excess supply of labor in the labor market, then market mechanism will cause the price of labor (wages) to fall and labor demand is increased by the firms and clears the market.



□ A negative demand shock decreases the labor demand and shifts the labor demand curve downward. If market mechanism works freely then wages would fall to W' and demand for labor will rise and shifts back again to D_L^* , and equilibrium is reestablished again at full employment level. But classical economists face the problem that wages do not fall in accordance with the labor demand. The reason why the wage rate do not fall much is that wages are sticky downwards. Wages get stuck to a level and do not fall below certain level.

(B) LABOR MARKET: THE KEYNESIAN VIEW

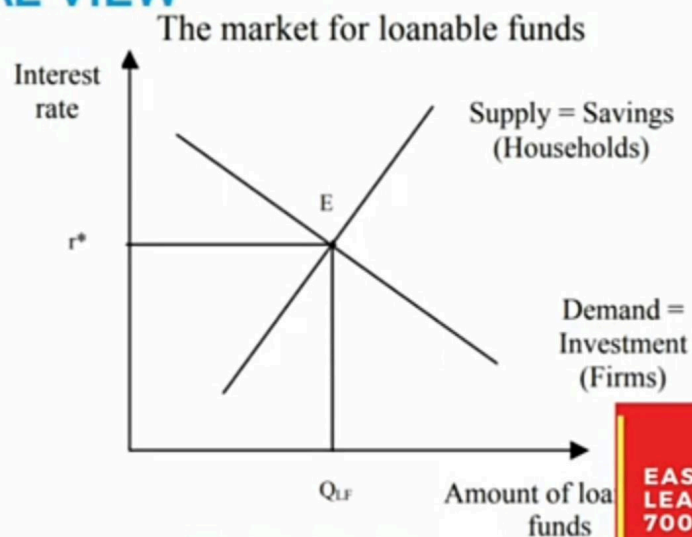
Keynes said once there is an excess supply of labor in the labor market and there is the pressure of wage rate to fall then firms look at the pressure of wage rate falling in a negative way. They think people are becoming poorer due to lower wages and they will not buy our products. So firms have no incentives to invest in the production of new goods.



□ If demand for labor falls to D_L' , then this would require wage rate to fall but once the wages fall then what would be the impact of this on the firms? Wages of consumers are used as the consumer spending. So when firms saw that people are becoming poorer they will have less incentive to produce more goods. So, investment falls and demand for labor also falls and further shifts downward to D_L'' . Wages would also decrease further. This will continue and increase unemployment. This is how Keynes explains the reasons of high unemployment in the period of great depression.

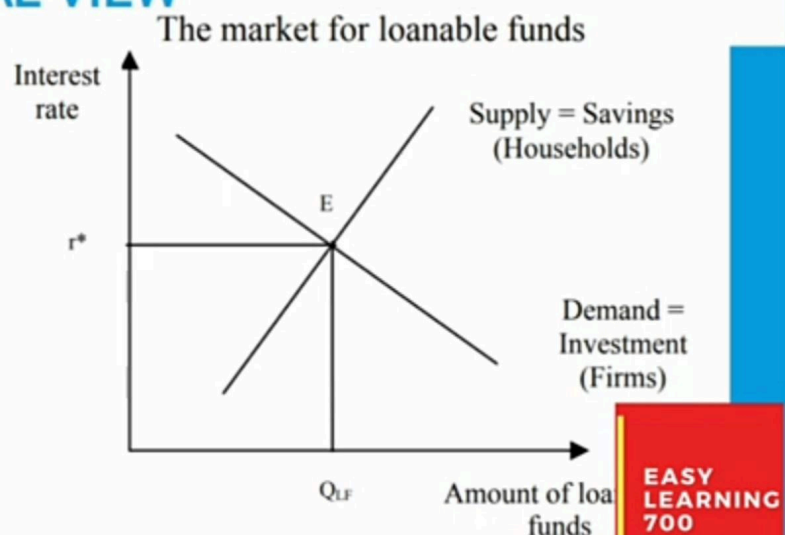
2- (A) MARKET FOR LOANABLE FUNDS: THE CLASSICAL VIEW

This is the market of money. Depositors provide the supply of loanable funds and business/ firms borrow the money generating the demand for loanable funds.

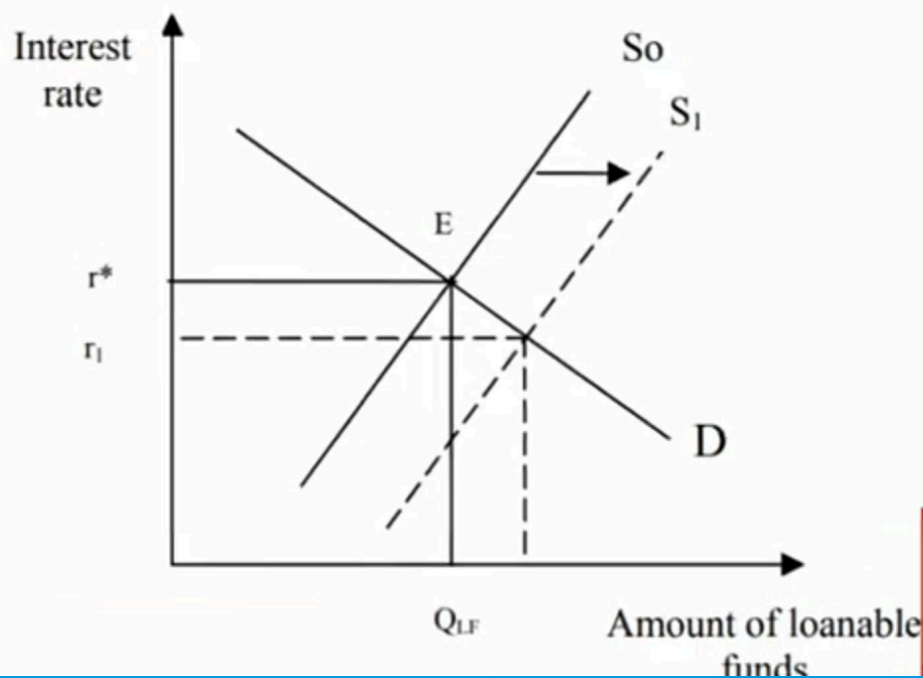


2- (A) MARKET FOR LOANABLE FUNDS: THE CLASSICAL VIEW

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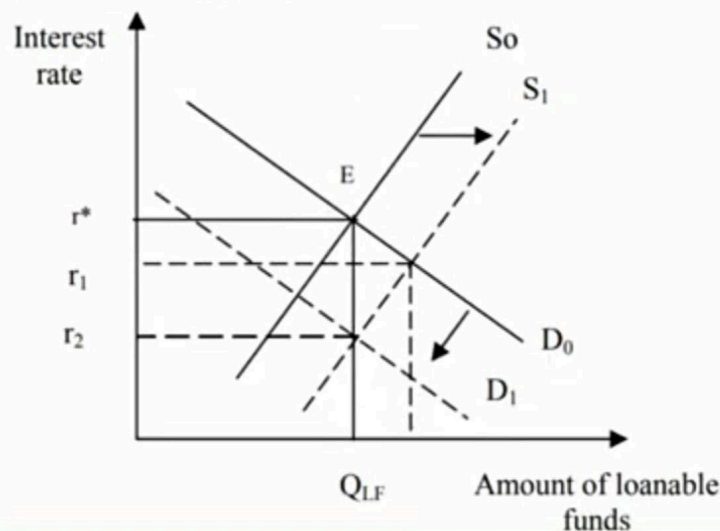
□ According to classicals, firms were not investing during great depression because of higher interest rate. Interest rate is the cost of borrowing the funds for the firms. When firms borrow funds, it has to pay the interest rate, that's why firm's investment was declined. According to them, if we lowered the interest rate then we can encourage the firms for investment. So individuals should save more, provide more and more money to the banks so that the supply of funds increase and interest rate fall.



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(B) MARKET FOR LOANABLE FUNDS: THE KEYNESIAN VIEW



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AGGREGATE DEMAND

- Aggregate demand (AD) is the total planned or desired spending (expenditure) in the economy during a given period. AD is the sum of consumption, investment, government spending and net exports (i.e. exports minus imports), and is inversely related to the aggregate price level through the wealth, interest rate and international purchasing power effects.
- AD curve slopes downward for both Keynes and classicals.

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WHY AD CURVE SLOPES DOWNWARD

AD curve slopes downward due to the following three effects.

1. The interest rate effect of a price level increase on AD is negative as it causes a fall in investment demand.
2. The wealth effect of a price level increase on AD is negative and works through the reduction in the purchasing power of consumers' income and wealth (real asset values).
3. The international purchasing power (or competitiveness) effect of a price level increase on AD is also negative as it reduces the net foreign demand for domestic goods and services.

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FACTORS THAT SHIFTS AGGREGATE DEMAND

AD shifts to the right when any component of AD increases autonomously; e.g., if

- a) Consumers become more willing to spend at every price level;
- b) There are autonomous increases in investment due to better business prospects;
- c) The government spends more, or reduces taxes;
- d) Net exports rise at all prices (due to say an increase in the quality of domestic goods relative to foreign goods).

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INTRODUCTION TO MACROECONOMICS (CONTINUED)

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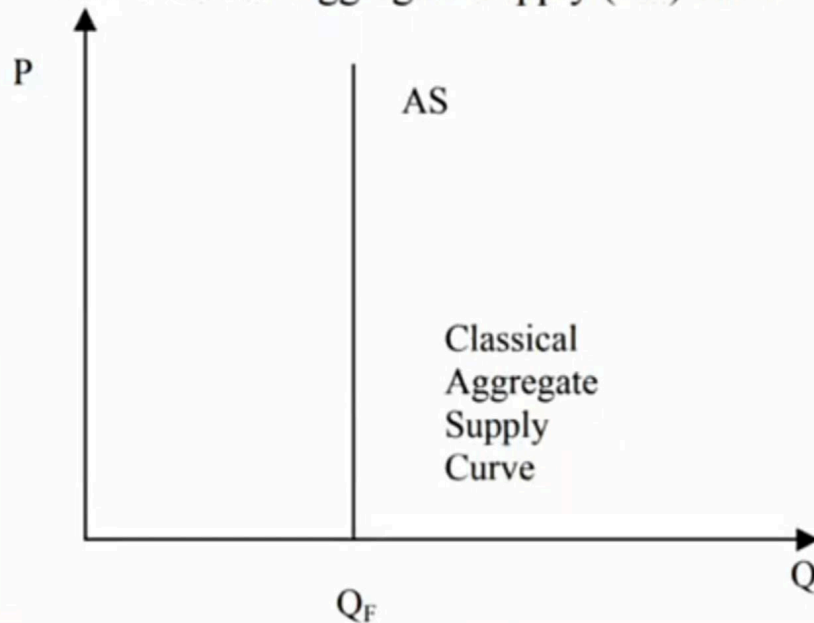
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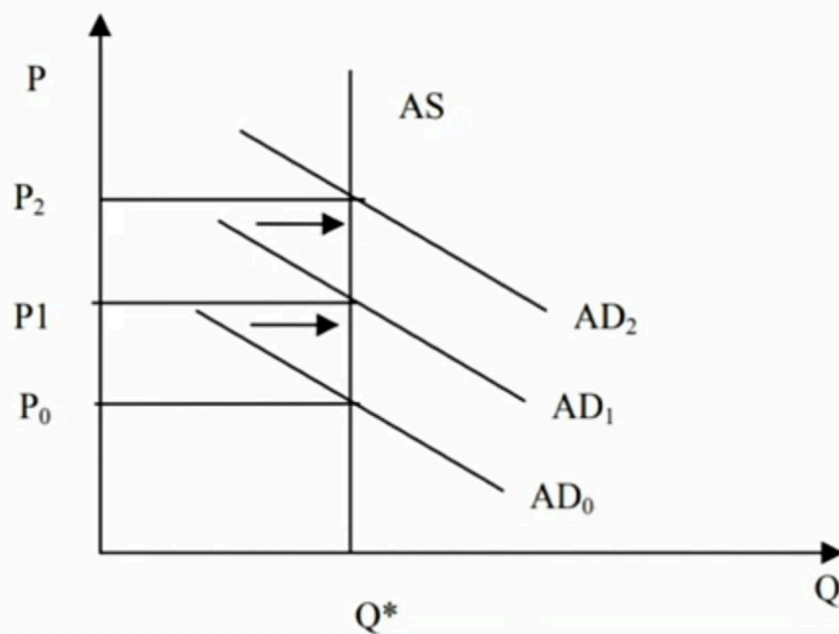
3- (A) AGGREGATE DEMAND AND SUPPLY: THE CLASSICAL VIEW

- The AS curve was vertical. According to classicals, economy is always at full employment level. Economy would automatically find the new equilibrium in the long run; they did not talk about short run.

The Classical Aggregate Supply (AS) curve



- Shifts in AD curve would have no effect on AS or on output level in the classical world. Any shift in AD curve will cause only change in the price level but output will not change. Output can change only if the AS curve would shift. AS curve can be shifted due to the availability of new resources, technology and wage rate.



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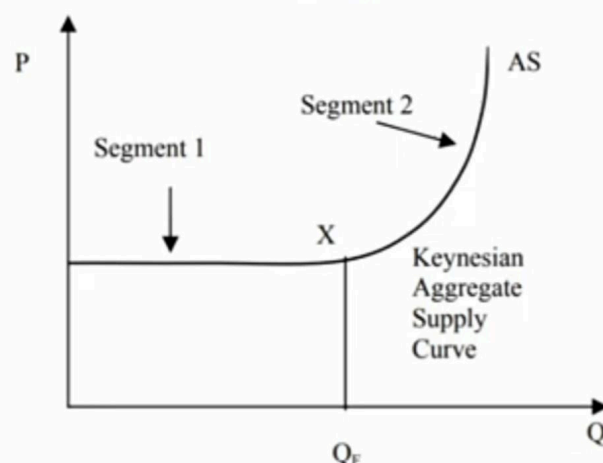
(B) AGGREGATE DEMAND AND SUPPLY: THE KEYNESIAN VIEW

The AS curve was horizontal at the less than full employment level (i.e. when there was excess capacity or slack in the economy), and upward sloping after that, Keynes said that output can be increased after increasing the price

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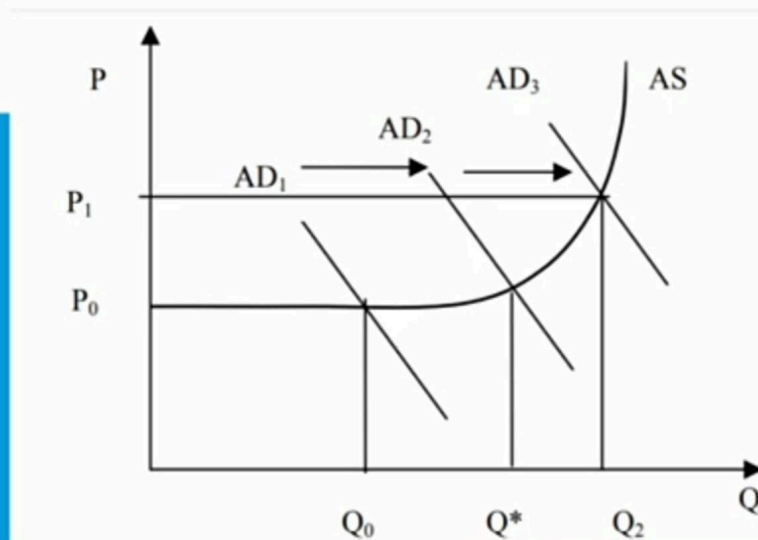
The Keynesian Aggregate Supply (AS) curve



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Shifts in AD curve would have the impact on the output level. Output will increase as the AD curve shifts rightward. Keynes said that prices are fixed in the short run.



KEYNES DEMAND MANAGEMENT POLICIES

Keynesian demand management policies were practiced by many governments when demand went "off" due to cyclical fluctuations of the economy. In recessions, the government increased spending and encouraged the private sector to do the same. In booms the opposite was done to cool the economy down.

Keynes' policies could not be applied in a situation where both prices and unemployment were rising (stagflation) – this situation arose in the 1970s with the two oil price shocks (which were essentially supply side shocks), and led to the decline of Keynesian economics.

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DIFFERENT SCHOOLS OF THOUGHTS

The Monetarist School:

The Monetarist School, led by Milton Friedman separated the explanation for inflation and unemployment. He noted that inflation was always and everywhere a monetary phenomenon and the key to keeping inflation low was to keep monetary growth aligned with expected real output growth.

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THE REAL BUSINESS CYCLES (RBC) SCHOOL:

The Real Business Cycles (RBC) School also gained currency in the 1970s. The exponents of the business cycles view noted that output fluctuated mainly due to technology shocks faced by the economy, and that no Keynesian type policy could, or should attempt to, neutralize their effects.

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THE RATIONAL EXPECTATIONS SCHOOL:

The same period, 1970s, saw the rise of the rational expectations school (as opposed to Keynes' static expectations hypothesis) led by such people as Robert Lucas, Robert Barro and Thomas Sargent who conceptualized agents as making use of all the information available to them, and not just past information, while making decisions. Under these and other conditions they showed that predictable macroeconomic policies (like Keynesian demand management policies) had no effect on real output or unemployment.

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NEO CLASSICAL ECONOMICS:

Coupled with the insights of the monetarist and business cycle schools, this view of the world reinforced the pre-Keynesian beliefs in the power of the free market and stressed the micro-foundations of macroeconomics. For this reason, it is called new or Neo Classical Economics.

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THE NEO KEYNESIAN SCHOOL:

Since the 1980s, the new or Neo Keynesian School has emerged, led by economists such as Joseph Stiglitz. The new Keynesians have highlighted market failures at the micro level that may arise due to information asymmetries and coordination failures (moral hazard and adverse selection problems). As such they have shown avenues for meaningful government intervention. This is broad where modern macroeconomics currently stands.

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Time Left 88 sec(s)

ECO401 - Economics (Quiz# 3)

Quiz Start Time: 09:57 PM, 25 July 2023

Question # 10 of 10 (start time: 10:07:54 PM, 25 July 2023)

Total Marks: 1

A decrease in the money supply will:



Reduce interest rates, reducing investment and GDP.



Raise interest rates, increasing investment and lowering GDP.



Reduce interest rates, increasing investment and GDP.



Raise interest rates, reducing investment and GDP.

Click to Save Answer & Move to Next Question



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Time Left 89 sec(s)

ECO401 - Economics (Quiz# 3)

Quiz Start Time: 09:57 PM, 25 July 2023

Question # 9 of 10 (start time: 10:06:33 PM, 25 July 2023)

Total Marks: 1

If fiscal deficit is financed through borrowing from the central bank, then the money supply will:

Select the correct option



Increase.



Decrease for short time then start increasing.



Remain unchanged.



Decrease.

Click to Save Answer & Move to Next Question

MACROECONOMIC DATA & NATIONAL INCOME ACCOUNTING

Lesson 28

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THE USE OF MACROECONOMIC DATA

As said: "there are lies, damned lies and statistics."
Likewise, macroeconomic statistics are also susceptible of both manipulation and misinterpretation.

In order to ensure that you understand what a particular number or data representation really means, the following need to be considered:

i. Data might be used selectively. e.g. inflation as a whole might have increased but food inflation might have fallen.

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ii. In graphs, the vertical and horizontal scales used might be such in a way so as to paint a very dramatic picture of things.

iii. Values used might be absolute, not proportionate.

iv. Questions of distribution might be ignored.

e.g. while the economy might have become richer overall. But richer have become richer and poor poorer.

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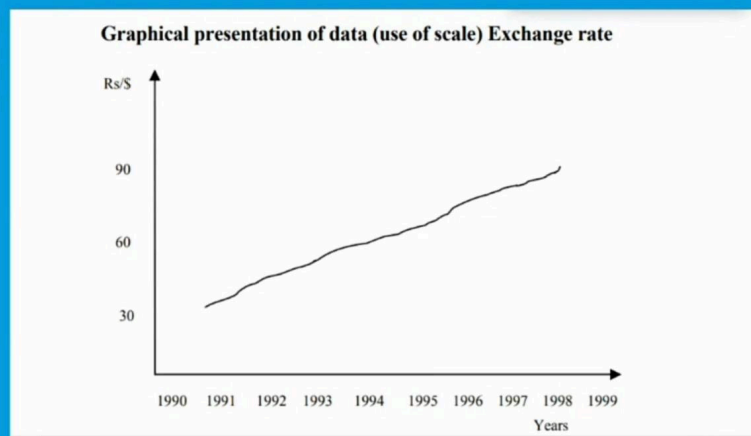
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viii. Vii also applies in the context of growth rates. So if national income is growing at 5% p.a. but the population at 6% p.a., per capital income would be falling at the rate of 1% p.a.

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5 people in the economy	Income in (000)s				
	A	B	C	D	E
Old government	100	110	120	130	140
New government	60	70	75	300	450

$$\text{Mean (old government)} = \frac{610}{5} = 122$$

$$\text{Median (old government)} = 120$$

$$\text{Mean (new government)} = \frac{955}{5} = 191$$

$$\text{Median (new government)} = 75$$

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NATIONAL INCOME ACCOUNTING

Gross Domestic Product (GDP):

Gross domestic product (GDP) is the value of the total final output produced inside a country, during a given year.

Concept of Flow and Stock:

Changes in stocks equal flows

Stock:

Stock: A variable or measurement that is defined for an instant in time (as opposed to a period of time). A stock can only be measured at a specific point in time.

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Flow:

A variable or measurement that is defined for a period of time (as opposed to an instant in time). A flow can only be measured over a period

METHODS OF MEASURING GDP

3 equivalent ways of measuring GDP:

i. The product or value added method:

which sums the value added by all the productive entities in the economy;

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ii. The expenditure method:

which sums up the value of all the “final goods” transactions taking place in the economy;

iii. The factor income method:

which sums up all the incomes earned by all the factors of production in the economy

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Value added:

Difference between the value of goods produced and the cost of materials and supplies used in producing them.

OR

Value added is the increase in the value of a good at each stage of the production process

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FINAL AND INTERMEDIATE GOODS:

Final goods are meant for direct use by the end consumer rather than for further processing. Intermediate goods are those that are intended for further processing

Final Good:

A good (or service) that is available for purchase by the ultimate or intended user with no plans for further physical transformation or as an input in the production of other goods that will be resold.

Intermediate good:

A good (or service) that is used as an input or component in the production of another good.

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DIFFERENCE BETWEEN TOTAL VALUE AND VALUE ADDED:

- Firm A: Produces steel from raw iron and sells it for Rs 100,000 to firm B.
- Firm B: Buys steel worth Rs 100,000 then processes it to produce a car body worth Rs 200,000.
- Firm C: Buys the car body, adds the other parts etc. and sells complete car for Rs 450,000.

□ Value added or GDP:

- Value of transactions: $100,000 + 200,000 + 450,000 = 750,000$ GDP = Sum of the value added by each of the firms: $= 100,000 + (200,000 - 100,000) + (450,000 - 200,000) = 100,000 + 100,000 + 250,000 = 450,000$ ∴ GDP = total expenditure of consumer on car = 450,000

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Example:

<u>Good</u> Col 1	<u>Seller</u> Col 2	<u>Buyer</u> Col 3	<u>Trans Value</u> Col 4
<u>Steel</u>	<u>Steel producer</u>	<u>Machine producer</u>	<u>100,000</u>
Steel	Steel producer	Car producer	300,000
Machine	Machine producer	Car producer	200,000
Tyres	Tyre producer	Car producer	50,000
Cars	Car producer	Consumer	500,000
Total value of transaction	-	-	1,150,000

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MACROECONOMIC DATA & NATIONAL INCOME ACCOUNTING (CONTINUED)

LECTURE 29

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GDP AT FACTOR AND MARKET PRICES:

Factor price is the price at which firm sells its final output to the consumers. While market price include factor price plus the indirect taxes imposed by the government.

$$\text{GDP at factor cost} = \text{GDP at market price} - \text{Indirect taxes}$$

Net Domestic Product (NDP):

Net domestic product (NDP) is obtained by subtracting depreciation from GDP.

$$\text{NDP} = \text{GDP} - \text{Depreciation allowance}$$

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Depreciation:

Wearing out, breaking down and technological obsolescence of physical capital that results by using in process of production of goods and services.

Gross National Product (GNP):

Gross national product (GNP) is the value, at current market prices, of all final goods and services produced during a year the factors owned by the citizens of a country.

$$\text{GNP} = \text{GDP} + \text{Net factor incomes from abroad}$$

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Net National Product (NNP):

Mathematically, national income is net national product (NNP). It is GNP adjusted for depreciation.

$$\text{NNP} = \text{GNP} - \text{Depreciation allowance}$$

National income (NI):

NNP is often referred to as national income.

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National Income and Gross Domestic Product:

National income (NI) is the total income earned by the citizens of the national economy resulting from their ownership of resources used in the production of final goods and services during a given period of time, usually one year.

Gross domestic product (GDP) is the total market value of all final goods and services produced within the political boundaries of an economy during a given period of time usually a year.

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Real GDP:

The total market value, measured in constant prices, of all goods and services produced within the political boundaries of an economy during a given period of time, usually one year.

Nominal GDP:

The total market value, measured in current prices, of goods and services produced within the political boundaries of an economy during a given period of time, usually one year.

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PRICE DEFLATOR:

The process of converting nominal GDP into real GDP is known as deflation.

$$\text{GDP Deflator} = \text{Nominal GDP} / \text{Real GDP}$$

Per Capita GDP:

Per capita GDP is simply the total GDP of the economy divided by the no. of people in the economy.

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THE PURCHASING PARITY (PPP) :

The purchasing power parity (PPP) measure of GDP recognizes the fact that a given amount of income in one country might not be able to purchase the same quantity of goods and services in another country.

e.g., if China's per capita income is 1/160th of Switzerland's per capita income, it might be that goods and services in China are much cheaper and therefore China's per capita income does not need to grow 160 times in order to deliver the same standard of living as in Switzerland.

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PER CAPITA INCOME, PERSONAL INCOME AND DISPOSABLE INCOME:

Per capita income

Per capita income is obtained by dividing the national income by the total number of population.

Personal income:

Personal income of an individual is the total amount of income s/he receives from deploying all the different factors of production s/he owns.

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Disposable income:

Disposable income is obtained by subtracting the amount of direct taxes from the personal income of the person.

DRAWBACK OF GDP BASED MEASURES:

i. GDP by definition excludes productive activities in the informal economy. Thus, activities such as a person painting a wall in his own house, or a woman cooking food in her house.

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ii. GDP cannot include the black or illegal economy. So, for example, if a banned good is produced illegally and exported outside the country, and foreign money is received in exchange for it.

iii. A GDP-based measure of welfare or living standards also needs to be corrected for externalities. For e.g., if a country's GDP is growing at a very fast rate but this is at the cost of rising environmental pollution (which might cause serious future health hazards) or non-renewable natural resource depletion, then a simple GDP measure of income will overstate the performance of the economy and ignore the serious long-term risks it faces.

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MACROECONOMIC EQUILIBRIUM; THE DETERMINATION OF EQUILIBRIUM INCOME

Lesson 30

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THE KEY VARIABLES OF THE MACROECONOMIC MODEL

The key variables for a macroeconomic model are:

Y = Income C = Consumption S = Savings I = Investment T = Taxes G = Government Expenditures M = Imports X = Exports

CIRCULAR FLOW:

Circular flow refers to the continuous movement of production, income, and resources between producers and consumers.

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THE CONCEPT OF LEAKAGES AND INJECTIONS

A leakage or withdrawal is any use of the income received by households that does not return as revenue to domestic firms.

Injectors are payments to firms not originating from households: government spending, firms' investment and exports are all examples of injections into the circular flow.

Injection:

A non-consumption expenditure on gross domestic product, including investment expenditures, government purchases, and exports.

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Leakage/Withdrawals:

Leakage is known as non-consumption uses of income, including saving, taxes, and imports.

Injection-leakage model:

A model used in Keynesian economics based on the equality of non-consumption expenditures (or injections) and non-consumption uses of income (leakages). On one side of the equality is saving, taxes, and imports -- the non-consumption leakages. On the other side of the equality is investment, government purchases, and exports -- the non-consumption injections. The injection-leakage model provides an alternative to the Keynesian cross for identifying equilibrium aggregate output.

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MACROECONOMIC EQUILIBRIUM: CLASSICAL VIEW

Macroeconomic equilibrium in a Classical sense refers to joint equilibrium in all the underlying sectors or markets of the economy. So S must equal I (loanable funds market; key players are banks and financial markets), G must equal T (fiscal sector; key player is government) and $X = M$ (external sector, key players are importers and exporters). Any disequilibrium at the macro level was attributable to disequilibrium in one or more of these individual markets.

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MACROECONOMIC EQUILIBRIUM: KEYNESIAN VIEW

Macroeconomic equilibrium in a Keynesian sense obtains when total injections equal total leakages (or total withdrawals), or aggregate supply equals aggregate demand. These are two equivalent notions of Keynesian equilibrium and can be expressed respectively as:

$$S + T + M \equiv I + G + X \text{ and}$$

$$AS = Y = AD \equiv C + I + G + (X - M);$$

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$$M = C_f + I_f + G_f$$

$$C_d = C - C_f$$

$$I_d = I - I_f$$

$$G_d = G - G_f$$

$$R.H.S = (C - C_f) + (I - I_f) + (G - G_f) + X$$

$$= C_d + I_d + G_d + X \Rightarrow \text{Total expenditure on domestic goods}$$

$$C_d + I_d + G_d + X = C + I + G + X - M$$

$$\text{Total expenditure on domestic goods} = AD$$

For Equilibrium

$$AD = AS$$

$$C + S + T = C + I + G + X - M$$

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In L.H.S

$$C + S + T = Y \text{ (total income} = Y)$$

Therefore another way of viewing
Keynesian equilibrium is

$$Y = AD = AS$$

$$\text{Income} = \text{Expenditure} = \text{Output}$$

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Disposable income (Y_d):

Disposable income (Y_d) is that part of the total national income (Y) that is available to households for consumption or saving. So $Y_d = Y - T$.

CONSUMPTION AND CONSUMPTION FUNCTION

Consumption (C) is the amount of national income that is spent on goods and services produced by domestic firms in a given period of time.

The consumption function is a schedule relating total consumption to personal disposable income. It usually takes the form $C = a + bY_d$

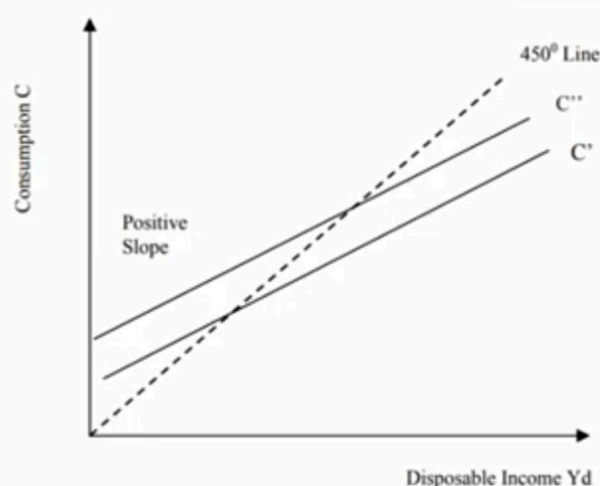
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where "a" is the minimum level of consumption that must take place even if Y_d is zero, and b is the marginal propensity to consume.

When drawn in expenditure-income space, the consumption function plots as a straight line with positive intercept, and a positive (but less than 1) slope. The slope is merely the MPC. The intercept is positive because some consumption must happen even at a zero level of income (people will buy food and spend on food for e.g.), and the slope is less than 1 because not all the income is consumed (part of it is saved).

Disposable Income (Y_d)	Consumption (C')	Saving ($S = Y_d - C'$)
500	500	0
550	540	10
600	580	20
650	620	30
700	660	40
750	700	50
800	740	60



MARGINAL PROPENSITY TO CONSUME (MPC) AND MARGINAL PROPENSITY TO SAVE (MPS):

- Marginal propensity to consume (MPC) is the extra amount that people consume when they receive an extra dollar of disposable income.
- Marginal propensity to save (MPS) is the fraction of the additional dollar of disposable income that is saved. Thus, $MPC = 1 - MPS$

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THE SAVING FUNCTION

The saving function yields the amount of saving that households of a nation will undertake at each level of income. A usual formula is $S = c + d(Y_d)$. d is MPS, positive, and usually less than 0.5. The relationship between saving and the interest rate is also important. The relationship is positive, is plotted in i - S space, and implies that household saving increases as the interest rate goes up, i.e. the incentive to keep one's money in the bank and earn interest thereon increases as the return on that money increases.

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- Average propensity to consume:

$$APC = C / Y_d$$

- Average propensity to save:

$$APS = S / Y_d$$

$$APS = 1 - APC$$

When $Y_d = \$500$ billion $APC = 1$ $APS = 0$

When $Y_d > \$500$ billion $APC < 1$ $APS > 0$

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- Marginal propensity to consume:

$$MPC = \Delta C / \Delta Y_d$$

- Marginal propensity to save:

$$MPS = \Delta S / \Delta Y_d$$

Or

$$MPS = 1 - MPC$$

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APC = C/Y _d	APS	Y _d	C	MPC(ΔC/ΔY _d)
500/500=1.0	0	500	500	
540/500=0.98	0.02	550	540	40/50=0.8
580/600=0.97	0.03	600	580	40/50=0.8
620/650=0.95	0.05	650	620	40/50=0.8
660/700=0.94	0.06	700	660	40/50=0.8
700/750=0.93	0.07	750	700	40/50=0.8
740/800=0.92	0.08	800	740	40/50=0.8

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